SMART WALKING STICK FOR VISUALLY IMPAIRED NAVIGATION WITH AI INTEGRATION

25-26J-270

Project Proposal Report

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Sri Lanka Institute of Information Technology
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DECLARATION

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We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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(Dr. Junius Anjana)

ABSTRACT

Globally, more than 2.2 billion people experience visual impairment, creating significant challenges in social participation and interpreting interpersonal cues. Traditional aids like white canes provide limited support when it comes to recognizing non-verbal signals such as facial expressions, age indicators, or gender traits—elements that are crucial for effective communication and social awareness. This part of the Smart Walking Stick project addresses the need for advanced facial analysis to enhance social perception and interaction for individuals with visual impairments.

The proposed system integrates ESP32-CAM modules with TinyML (TensorFlow Lite) models to enable real-time recognition of emotions (happy, sad, angry, along with recommended additions like neutral, surprised, fear, and disgust), age group estimation (e.g., 20–25, 30–40, 40–50, 50+), and gender classification (male or female). The ESP32-CAM captures facial images, processes them through a lightweight model for multi-attribute classification, and delivers instant feedback using haptic vibrations and audio alerts. The emphasis is on detecting essential facial attributes of nearby individuals to foster better social interaction.

By combining computer vision, on-device processing, and user-centered interface design, this system ensures efficient, low-latency, and energy-conscious performance with reliable accuracy in attribute detection, making it suitable for embedding in a portable assistive device. Expected outcomes include improved social interactions, greater awareness of interpersonal details, and increased confidence in everyday communication for people with visual impairments.

Keywords: Facial Emotion Recognition, Computer Vision, TinyML, ESP32-CAM, Assistive Technology, Visual Impairment

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
CNN	Convolutional Neural Network
ETA	Electronic Travel Aid
FPS	Frames Per Second
IoT	Internet of Things
MTBF	Mean Time Between Failures
PCB	Printed Circuit Board
Tiny ML	Tiny Machine Learning
FR	Functional Requirements
NFR	Non-Functional Requirement

1. INTRODUCTION

Visual impairment presents a significant obstacle to social connection and emotional awareness for those affected. According to the World Health Organization, nearly 2.2 billion people worldwide live with some form of vision loss, with at least 1 billion cases being either preventable or untreated [1]. Individuals with limited vision often face difficulties in interpreting facial expressions, estimating age ranges, or recognizing gender during interactions. These challenges can lead to miscommunication, social isolation, and reduced confidence in group environments.

Conventional mobility aids, such as white canes or guide dogs, primarily support navigation but fail to convey nuanced social signals like emotions or demographic characteristics [2]. While service animals may provide companionship, they cannot communicate specific facial details. With recent progress in IoT, computer vision, and artificial intelligence, smarter assistive technologies are now becoming feasible to bridge this gap.

Facial attribute recognition plays a crucial role in human communication, as non-verbal elements are estimated to account for up to 93% of emotional expression [3]. For individuals with vision loss, the inability to perceive these cues can hinder relationships, limit professional opportunities, and weaken everyday interactions. Current assistive solutions focus on mobility rather than providing social-emotional support.

Integrating visual analysis with compact hardware enables systems capable of detecting, classifying, and relaying facial attributes in real time. The ESP32-CAM, with its built-in camera and wireless connectivity, is an ideal choice due to its efficiency, portability, and compatibility with lightweight AI models [4].

This research proposes a facial attribute recognition system built on ESP32-CAM and TensorFlow Lite that identifies emotions (happy, sad, angry, neutral, surprised, fear, disgust), estimates age groups (e.g., 20–25, 30–40), determines gender (male/female), and provides immediate feedback through vibrations and audio alerts. The approach enhances assistive technology by combining edge AI, computer vision, and inclusive design into a cost-effective tool for improving social awareness.

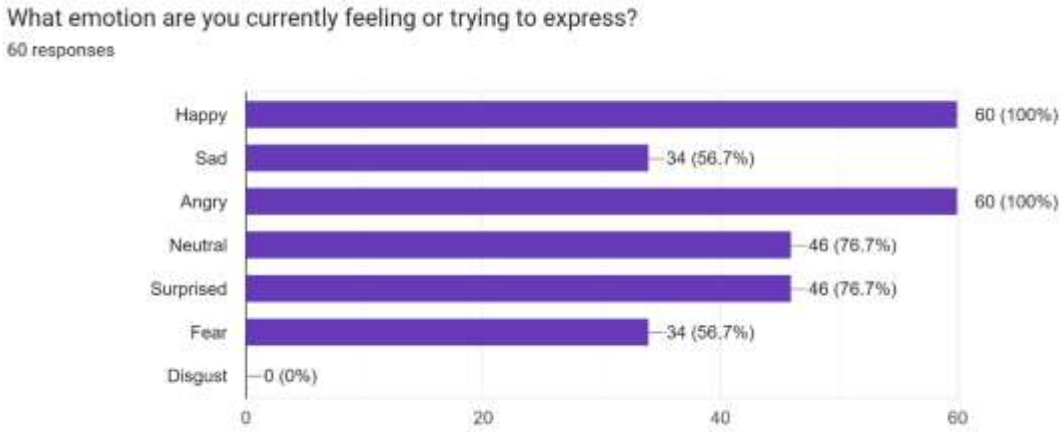

Figure 1.1 What emotion people currently feeling or trying to express

1.1 Background & Literature Survey

1.1.1 Assistive technologies for visual impairment

Assistive tech for the visually impaired has advanced from basic tools to AI-enhanced devices. White canes from the 1930s detect obstacles but ignore social cues [5]. Early electronic aids like ultrasonic sensors expanded detection but centered on barriers [6]. Contemporary systems add AI, yet emotional and demographic analysis is sparse. Sound-based aids exist, but they bypass visual social signals [7].

1.1.2 Computer vision in assistive technology

Advancements in vision-based technologies have transformed assistive tools, enabling features such as text reading and object detection [8]. When it comes to facial attributes, convolutional neural networks (CNNs) demonstrate strong performance in recognizing emotions, estimating age, and classifying gender [9]. Lightweight

architectures like MobileNet make it possible to deploy these models on edge devices, ensuring real-time analysis and responsiveness [10].

1.1.3 Edge computing and TinyML

On-device processing minimizes latency, which is crucial for maintaining natural social interactions. TinyML enables machine learning models to operate directly on microcontrollers, while TensorFlow Lite provides optimization for devices with limited resources [11]. Such capabilities make the technology well-suited for integration into wearable and assistive tools [12].

1.1.4 ESP32-CAM in computer vision applications

Moreover, its Wi-Fi and Bluetooth capabilities enable seamless data transmission, making it highly suitable for Internet of Things (IoT)–based facial recognition systems [15]. The integration of edge computing with ESP32-CAM reduces dependency on cloud servers, ensuring faster processing and improved privacy [16]. Additionally, its adaptability allows deployment in various scenarios such as smart surveillance, access control, and assistive technologies for visually impaired individuals [17].

1.1.5 Facial attribute recognition systems

Existing landmark recognition systems have primarily focused on large-scale applications such as autonomous vehicles and augmented reality. These systems often rely on extensive computational resources and high-resolution cameras, making them unsuitable for portable assistive devices [15].

Some research has been conducted on landmark recognition for navigation assistance, but most solutions require smartphone-based processing or cloud connectivity, which may not be practical for real-time applications or users with limited access to modern smartphones [16].

1.1.6 Human computer interaction in assistive technology

Recent studies also explore multimodal feedback systems, combining haptic, audio, and even tactile displays to enhance user experience and reduce cognitive load [19].

Such systems can adapt feedback intensity or type based on contextual factors, like distance or the number of detected faces, to improve situational awareness [20]. Additionally, integrating machine learning–based prediction allows these interfaces to provide personalized guidance, increasing both accuracy and user confidence in real-time applications [21].

1.2 Research gap

Despite advancements in AI and attribute recognition, significant challenges remain for visually impaired individuals. Many current technologies still struggle to accurately interpret complex visual environments. As a result, these gaps limit accessibility and independence for those who rely on assistive tools.

1.2.1 Limited real-time processing capabilities

Many systems use external processing, causing delays unfit for social timing. They favor precision over immediacy. In real-world interactions, even small lags can make communication feel unnatural or isolating. This creates frustration for visually impaired users who depend on quick feedback to navigate their surroundings. While accuracy is important, responsiveness is equally critical for seamless daily experiences.

1.2.2 Lack of embedded, portable solutions

Tech often needs extra gadgets, raising costs. Aid integration is hindered by power needs. Frequent charging requirements and short battery life reduce reliability for daily use. For many users, carrying multiple devices adds both physical and financial burden. These barriers limit widespread adoption, even when the core technology shows strong potential.

1.2.3 Insufficient focus on multi-attributes

Focus is on basics, ignoring combined emotion, age, and gender for social depth. Most systems detect objects or simple attributes but miss the subtle cues that shape human interaction. Without recognizing emotional states or social context, assistive tools remain limited in supporting natural communication. This lack of depth reduces inclusivity, making social experiences less meaningful for visually impaired individuals.

1.2.4 Limited customization and adaptability

Models use standard data, missing cultural or personal tweaks. They often fail to capture regional expressions, gestures, or dress codes that hold social meaning. This one-size-fits-all approach overlooks individual preferences and cultural diversity. As a result, the assistance provided may feel generic or even inaccurate in real-world contexts.

1.2.5 Inadequate feedback mechanisms

Audio-only fails in noise; combined haptic-audio is rare. In crowded or loud environments, spoken feedback becomes hard to follow, reducing effectiveness. Haptic signals could provide silent, immediate guidance, yet they remain underused in most assistive tools. A balanced multimodal approach could greatly improve reliability and accessibility in real-life scenarios.

1.2.6 Cost and accessibility barriers

High costs limit reach in varied regions. Advanced assistive technologies often remain accessible only to users in wealthier communities. In low-resource settings, affordability becomes a major barrier, leaving many visually impaired individuals without access to essential tools. This gap reinforces inequality and slows global adoption of inclusive solutions

.

Table 1.1 below provides a comparison of existing smart stick systems with the proposed system, highlighting how current solutions address or fail to address these critical gaps.

Table 1.1 comparison of existing smart stick systems with the proposed system

Feature / System	We WALK Smart Cane (Research 1)	Smart Cane IIT Delhi (Research 2)	IoT-Based Smart Stick (Research 3)	A IoT Smart Stick (Research 4)	Proposed Research
Real-time Facial Attribute Recognition	✕	✕	✕	✕	✓
Embedded Processing	✕	✓	✓	✕	✓
Multi-Attribute Detection (Emotion/Age/Gender)	✕	✕	✕	✕	✓
Multimodal Feedback (audio + haptic)	✓ (audio only)	✓ (vibration only)	✓ (audio only)	✓ (audio only)	✓ (audio + haptic)
Portable / Integrated	✓	✓	✓	✓	✓
Affordable / Accessible	✕	✓	✓	✕	✓

1.3 Research problem ====

Visually impaired face hurdles in decoding facial attributes, causing social gaps. Aids ignore emotions (happy, sad, angry, etc.), age, and gender, leading to missed nuances in interaction. Without these cues, conversations may feel incomplete or disconnected. This limitation reduces confidence in social engagement and restricts deeper relationship building. Integrating richer attribute recognition could bridge these gaps and create more natural communication experiences.

1.3.1 Social uncertainty

Without the ability to accurately recognize facial attributes such as moods, age, and gender, visually impaired individuals often face uncertainty in social interactions and spatial awareness. This limitation can make it difficult to interpret conversations, assess social context, and gauge the presence or orientation of others in their environment. As a result, users may experience awkward or uncomfortable interactions, increased reliance on assistance, and reduced confidence in both navigation and social engagement. Reliable recognition of these subtle cues is therefore essential for fostering trust, independence, and smoother human connections.

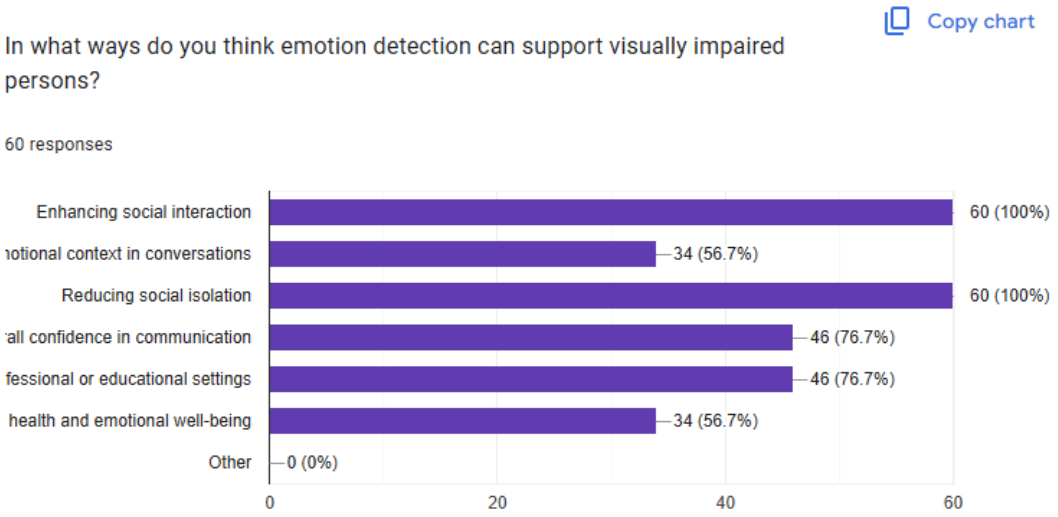

Figure 1.2 In what ways do you think emotion detection can support visually impaired people?

1.3.2 Safety concerns

Missing emotions like anger or fear, or misjudging demographics, risks harm for visually impaired users. Misinterpreting social cues can lead to unsafe situations, misunderstandings, or inappropriate responses. Without accurate recognition, users may struggle to navigate crowded or potentially hostile environments safely. Ensuring

precise detection of emotional and demographic cues is therefore critical for both personal safety and effective social interaction.

1.3.3 Limited environmental awareness

No grasp of combined cues hampers bonds. When assistive tools fail to integrate emotions, age, gender, and contextual signals, users miss the subtle dynamics that shape meaningful relationships. This can result in shallow interactions, misunderstandings, and reduced trust. Over time, the inability to perceive these layered social cues may limit social confidence, hinder community participation, and prevent the formation of deeper personal connections.

1.3.4 Technology limitations

Reliance on networks, processing delays, and costly setups limits the practicality of current assistive technologies. Dependence on cloud-based processing can create lag, making real-time interaction difficult. Expensive hardware and specialized devices reduce accessibility, especially for users in low-resource settings. Together, these factors restrict widespread adoption and reduce the overall effectiveness of tools designed to aid visually impaired individuals in daily life.

1.3.5 User experience challenges

Current systems often fail to provide intuitive and accessible user interfaces, with complex setup and operation procedures, overwhelming or unclear feedback mechanisms, limited customization options for different user needs, and inadequate training and support resources.

The research problem can be formulated as follows:

Problem: How to build an embedded facial attribute system for real-time emotion (happy, sad, angry, neutral, surprised, fear, disgust), age (e.g., 20–25, 30–40), and gender (male/female) detection to improve social awareness for visually impaired individuals, while tackling cost, mobility, and access issues? Current solutions often rely on cloud-based processing, leading to delays, privacy concerns, and dependence on stable internet connections. Many existing systems also require expensive hardware or multiple external devices, making them less practical for everyday use. An effective solution must operate efficiently on lightweight, portable, and affordable embedded

platforms, ensuring real-time performance, accessibility, and reliability in diverse social and environmental contexts.

2. OBJECTIVES

2.1 Main objective

To create an advanced facial attribute recognition system via ESP32-CAM and TinyML for live detection/feedback of emotions (happy, sad, angry, neutral, surprised, fear, disgust), age groups (e.g., 20-25, 30-40, 40-50, 50+), gender (male/female) to boost social awareness for visually impaired.

2.2 Specific objectives

2.2.1 Machine learning model development

- Train/optimize TensorFlow Lite model for multi-attribute classification with ≥85% accuracy
- Use quantization for size/performance balance
- Build dataset from FER-2013, UTKFace for emotions, age, gender
- Test under diverse lights/distances

2.1.2 ESP32-CAM implementation

- Program ESP32-CAM for live face capture and preprocessing
- Implement TinyML inference capabilities on ESP32-CAM hardware
- Tune camera for attribute spotting
- Optimize pipelines for low latency

2.2.3 Communication system integration

- Set serial links to Arduino Uno
- Transmit attribute data reliably
- Ensure quick, error-free comms
- Add recovery protocols

2.2.4 Feedback system development

- Craft haptic patterns for attributes (e.g., short vibe for happy/male/20-25)
- Build customizable audio cues
- Map outputs for clear distinction
- Validate with users

2.2.5 System integration and testing

- Merge with walking stick
- Run field tests
- Measure accuracy, speed, reliability
- Do acceptance trials

2.2.6 Performance optimization

- Cut power for long runtime
- Reduce delay for instant feedback
- Build weather resilience
- Add adaptive improvements

3. METHODOLOGY

3.1 Overall system description

The facial attribute component enhances the Smart Walking Stick by integrating social awareness features into mobility assistance. Using an ESP32-CAM module, the system captures facial images in real time and leverages TinyML models to classify emotions, age groups, and gender directly on the device. This embedded approach minimizes dependence on external networks, reducing delays and improving privacy. Detected attributes are then translated into intuitive haptic or audio feedback, enabling visually impaired users to better understand the social context of their surroundings. By combining navigation support with social interaction cues, the Smart Walking Stick aims to improve both independence and confidence in daily life.

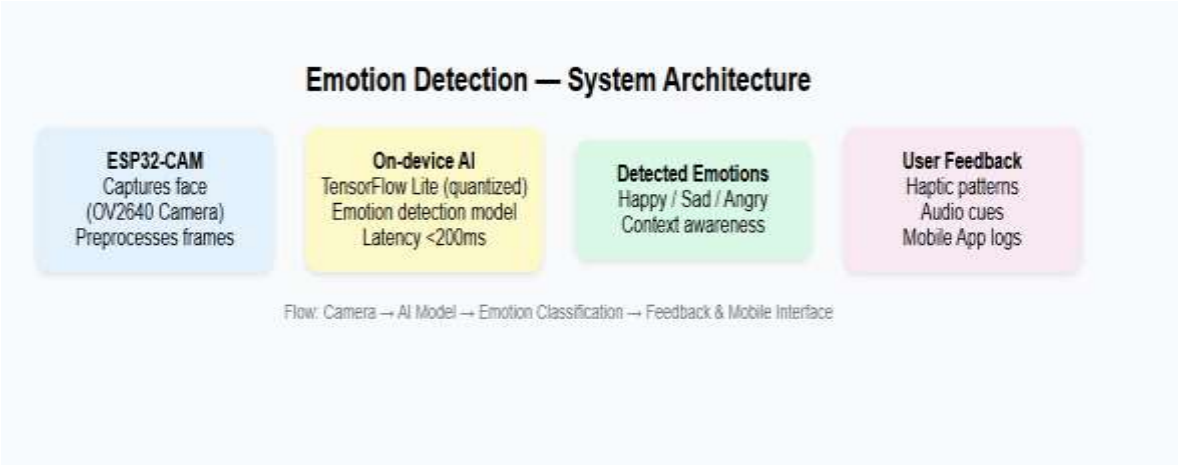

Figure 3.1 System Architectural Diagram

3.2 Development methodology

The project follows an iterative development approach based on the Rapid Application Development (RAD) methodology, allowing for quick prototyping and continuous refinement based on user feedback and testing results.

3.2.1 Ghantt Chart

The Gantt chart for the "Smart Walking Stick for Visually Impaired" project outlines a structured timeline from August 2025 to June 2026. It details key phases, including planning, design, and model development, with specific tasks such as project initiation, literature review, hardware/software architecture design, and mobile app development. Each task is color-coded and scheduled to ensure progress, quick prototyping, and continuous refinement based on user feedback and testing results.

Figure 3.2 Gantt Chart for the entire research project

Full hosted Ghantt chart for the whole year - https://ghant-chart-25-26j-270.netlify.app/

3.2.2 Work Breakdown Structure

Phase 1: Research and Planning (Weeks 1-4)

- Literature review and technology assessment
- Dataset collection and preparation
- Hardware component selection and procurement
- Development environment setup

Phase 2: Model Development (Weeks 5-12)

- Dataset creation and annotation
- Model architecture design and training
- Model optimization and quantization
- Performance evaluation and validation

Phase 3: Hardware Implementation (Weeks 13-20)

- ESP32-CAM programming and configuration
- Communication protocol implementation
- Hardware integration and testing
- Power optimization and reliability testing

Phase 4: System Integration (Weeks 21-28)

- Component integration with main system
- Feedback system development and testing
- User interface development
- Comprehensive system testing

Phase 5: Evaluation and Refinement (Weeks 29-36)

- Real-world testing and validation
- User acceptance testing and deployment
- Performance optimization and documentation

3.3 Technical implementation approach

3.3.1 Machine learning model development

Dataset Creation:

- Combine FER-2013/UTKFace with augments for variety
- Cover conditions
- Implement data augmentation techniques to increase dataset diversity
- Create balanced datasets with equal representation

Model Architecture:

- Utilize MobileNetV2 as the base architecture for efficient inference on embedded devices
- Implement transfer learning techniques using pre-trained ImageNet weights
- Tune params
- Apply quantization techniques to reduce model size and inference time

Training Process:

- Implement stratified data splitting to ensure representative training and validation sets
- Use cross-validation techniques to evaluate model robustness
- Apply regularization techniques to prevent overfitting
- Optimize hyperparameters using grid search and random search methods

3.3.2 ESP32-CAM implementation

Hardware Configuration:

- Configure ESP32-CAM camera settings for optimal image quality
- Implement automatic exposure and white balance adjustment
- Set up efficient memory management for image processing
- Configure power management settings for extended battery life

Software Development:

- Develop image capture and preprocessing modules
- Implement TensorFlow Lite inference engine
- Create communication protocols for data transmission
- Develop error handling and recovery mechanisms

3.3.3 Communication system

- Implement reliable serial communication between ESP32-CAM and Arduino
- Serial with integrity checks
- Implement acknowledgment mechanisms for data integrity
- Create timeout and retry mechanisms for robust communication

3.3.4 Feedback system design

Audio Alerts:

- Customizable tone frequencies for different landmark types
- Voice announcements for detailed landmark information
- Volume adjustment based on ambient noise levels
- Multiple language support

3.4 Testing and validation methodology

Laboratory testing

- Controlled environment testing
- Performance benchmarking under various lighting conditions
- Stress testing for system reliability and robustness
- Power consumption measurement and optimization

Field testing

- Real-world testing in urban environments
- Performance evaluation in different weather conditions
- User acceptance testing with visually impaired participants
- Comparative analysis with existing assistive technologies

Evaluation metrics

- Classification accuracy and precision
- System response time and latency
- Power consumption and battery life
- User satisfaction and usability scores

4. PROJECT REQUIREMENTS

4.1 Functional requirements

4.1.1 Core functionality

- **FR1:** The system shall capture high-resolution images using ESP32-CAM at minimum 2 frames per second
- **FR2:** Classify attributes ≥85% accuracy
- **FR3:** Spot emotions (happy/sad/angry/neutral/surprised/fear/disgust), age bins, gender
- **FR4:** The system shall provide real-time feedback within 500ms
- **FR5:** Send data serial to Arduino

4.1.2 User interface requirements

- **FR6:** The system shall provide distinct vibrotactile patterns for different landmark categories
- **FR7:** The system shall generate audio alerts with customizable volume levels
- **FR8:** The system shall support multiple feedback modes (vibration only, audio only, combined)
- **FR9:** The system shall allow users to enable/disable specific categories
- **FR10:** The system shall provide battery status indicators through the mobile application

4.1.3 Communication requirements

- **FR11:** The system shall maintain reliable serial communication with Arduino controller
- **FR12:** The system shall implement error detection and correction mechanisms
- **FR13:** The system shall support data transmission rates of minimum 9600 baud
- **FR14:** The system shall integrate with the mobile application via Bluetooth connectivity

4.1.4 Configuration requirement

- **FR15:** The system shall allow adjustment of detection sensitivity levels
- **FR16:** The system shall support multiple language options for audio feedback
- **FR17:** The system enables users to customize feedback patterns
- **FR18:** The system shall store user preferences in non-volatile memory

4.2 Non-functional requirements

4.2.1 Performance requirements

- **NFR1:** System response time shall not exceed 500ms from image capture to feedback delivery
- **NFR2:** The system shall achieve minimum 85% classification accuracy
- **NFR3:** The system shall operate continuously for minimum 8 hours on a single battery charge
- **NFR4:** The system shall process images at minimum 2 frames per second
- **NFR5:** Memory usage shall not exceed 80% of available ESP32-CAM memory

4.2.2 Reliability requirements

- **NFR6:** The system shall have an availability rate of 99.5% during normal operation

- **NFR7:** Mean time between failures (MTBF) shall exceed 1000 hours
- **NFR8:** The system shall recover from communication errors within 2 seconds
- **NFR9:** Data transmission error rate shall not exceed 0.1%
- **NFR10:** The system shall operate reliably in temperature range of -10°C to +50°C

4.2.3 Usability requirements

- **NFR11:** Learning time for new users shall not exceed 30 minutes
- **NFR12:** The system shall provide intuitive feedback that requires no prior training
- **NFR13:** Setup and configuration time shall not exceed 10 minutes
- **NFR14:** The system shall be operable by users with limited technical expertise
- **NFR15:** User interface elements shall comply with accessibility standards

4.2.4 Portability and integration requirements

- **NFR16:** The ESP32-CAM module shall weigh less than 50 grams
- **NFR17:** The system shall integrate seamlessly with the overall walking stick design
- **NFR18:** Power consumption shall not exceed 500mA during normal operation
- **NFR19:** The system shall be resistant to water splashing (IPX4 rating)
- **NFR20:** Operating range for Bluetooth communication shall be minimum 10 meters

4.3 System requirements

4.3.1 Hardware requirements

- **SR1:** ESP32-CAM development board with minimum 4MB flash memory
- **SR2:** OV2640 camera sensor with 2MP resolution capability
- **SR3:** Arduino Uno R3 or compatible microcontroller board

- **SR4:** HC-05 Bluetooth module for wireless communication
- **SR5:** Vibration motors with minimum 3V operating voltage
- **SR6:** Piezo buzzer for audio feedback generation
- **SR7:** Rechargeable lithium-ion battery with minimum 2000mAh capacity
- **SR8:** Voltage regulation circuits for stable power supply

4.3.2 Software requirements

- **SR9:** Arduino IDE version 1.8.19 or later for development
- **SR10:** ESP32 board package for Arduino IDE
- **SR11:** TensorFlow Lite for Microcontrollers library
- **SR12:** OpenCV library for image preprocessing
- **SR13:** Flutter framework for mobile application development
- **SR14:** Firebase SDK for cloud services integration

5. BUDGET AND BUDGET JUSTIFICATION

5.1 Hardware costs

Table 5.1: Hardware costs

Component	Qty	Unit Price (LKR)	Total Cost (LKR)	Justification
ESP32-CAM Development Board	2	2,100	4,200	Multiple units for testing, development, and backup
Arduino Uno R3	2	3,000	6,000	Primary and backup controllers
HC-05 Bluetooth Module	1	1,500	1,500	Communication with mobile device
Vibration Motors (3V)	5			Haptic feedback generation
Piezo Buzzer	3			Audio alert generation
Lithium-ion Battery (18650)	3			Power supply for system
Battery Holder and Charging Circuit	2			Safe battery management
Jumper Wires and Connectors	1 Set			System connectivity
Breadboards and PCB	3			Prototyping and final assembly
Enclosure and Mounting Hardware	1 Set			Weather Protection and integration
Hardware Subtotal				

5.2 Software and development costs

Table 2 Software and development cost

Item	Cost (LKR)	Justification
Arduino IDE and Libraries	Free	Open-source development environment
TensorFlow Lite	Free	Open-source machine learning framework
Image Dataset Creation		Data collection, annotation, and processing
Cloud Storage for Dataset	2,400	Google Drive Pro subscription for 6 months
Mobile App Development Tools	Free	Flutter framework is open source
Software Subtotal		

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